12 Continuous Random Variables

- A continuous random variable can take any value within some uncountable interval.
- Simulated values of a continuous random variable are usually plotted in a **histogram** which groups the observed values into "bins" and plots densities or frequencies for each bin.
- In a histogram *areas* of bars represent relative frequencies; the axis which represents the height of the bars is called "density".
- The continuous analog of a probability mass function (pmf) for a discrete random variable is a probability density function (pdf) However there are some fundamental differences between discrete and continuous random variables, and pmfs and pdfs.
- The distribution of a continuous random variable can be described by a **probability density function (pdf)**, for which *areas under the density curve determine probabilities.*

Example 12.1. Suppose that in your office everyone arrives to work between 8am and 9am. Randomly select an employee and let X be employee's arrival time measured in minutes after 8am, including fractions of a minute, so that X takes a value in the continuous interval [0, 60].

Data on many randomly selected employees suggests that the distribution of X is described by this probability density function (pdf), where c is a constant that we'll determine soon.

$$f_X(x) = \begin{cases} cx, & 0 < x < 60 \\ 0, & \text{otherwise.} \end{cases}$$

1. Sketch a plot of the pdf. Roughly, what does this say about arival times?

2. Find the value of c and specify the pdf of X.

3. Compute and interpret $\mathrm{P}(X < 15)$.

4. Compute and interpret $\mathrm{P}(X > 45)$.

- The continuous analog of a probability mass function (pmf) is a *probability density function (pdf)*.
- However, while pmfs and pdfs play analogous roles, they are different in one fundamental way; namely, a pmf outputs probabilities directly, while a pdf does not.
- A pdf of a continuous random variable must be *integrated* to find probabilities of related events.
- The **probability density function (pdf)** (a.k.a. density) of a *continuous* RV X is the function f_X which satisfies

$$\mathrm{P}(a \le X \le b) = \int_a^b f_X(x)dx, \qquad \text{for all } -\infty \le a \le b \le \infty \tag{12.1}$$

- For a continuous random variable X with pdf f_X, the probability that X takes a value in the interval $[a, b]$ is the *area under the pdf over the region* $[a, b]$.
- The axioms of probability imply that a valid pdf must satisfy

$$f_X(x) \ge 0 \qquad \text{for all } x, \tag{12.2}$$

$$\int_{-\infty}^{\infty} f_X(x)dx = 1 \tag{12.3}$$

Example 12.2. Continuing Example 12.1, recall that X has pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 \le x \le 60, \\ 0, & \text{otherwise.} \end{cases}$$

1. Consider the probability that an employee arrives within 1 minute after 8:15. Compute $\mathrm{P}(15 < X < 16)$, the probability that X, truncated to the nearest minute, is 15. How is the probability related to $f(15)$?

2. Consider the probability that an employee arrives within 1 minute after 8:45. Compute $P(45 < X < 46)$, the probability that X, truncated to the nearest minute, is 45. How is the probability related to $f(45)$?

3. Compute and interpret the ratio of the two previous probabilities.

4. Consider the probability that an employee arrives within 1 *second* after 8:15. Compute $P(15 < X < 15 + 1/60)$, the probability that X, truncated to the nearest *second*, is 15. (The measurement units of X are minutes, so 1/60 minutes is 1 second.) How is the probability related to $f(15)$?

5. Consider the probability that an employee arrives within 1 *second* after 8:45. Compute $P(45 < X < 45 + 1/60)$, the probability that X, truncated to the nearest *second*, is 45. How is the probability related to $f(45)$?

6. Compute and interpret the ratio of the two previous probabilities.

7. Consider the probability that an employee arrives within 1 *millisecond* after 8:15. Compute $P(15 < X < 15 + 1/60000)$, the probability that X, truncated to the nearest *millisecond*, is 15. (The measurement units of X are minutes, so 1/60000 minutes is 1 millisecond.) How is the probability related to $f(15)$?

8. Consider the probability that an employee arrives within 1 *millisecond* after 8:45. Compute $\mathrm{P}(45 < X < 45 + 1/60000)$, the probability that X, truncated to the nearest *millisecond*, is 45. How is the probability related to $f(45)$?

9. Compute and interpret the ratio of the two previous probabilities.

10. Compute and interpret $\mathrm{P}(X = 15)$.

11. Compute and interpret $\mathrm{P}(X = 45)$.

12. In practical terms, is any employee more likely to arrive "at 8:15" or "at 8:45"? Explain.

- **The probability that a continuous random variable X equals any particular value is 0**. That is, if X is continuous then $\mathrm{P}(X = x) = 0$ for all x.
- For a *continuous* random variable X, $\mathrm{P}(X \le x) = \mathrm{P}(X < x)$.
  - Be careful: for a *discrete* random variable X, $\mathrm{P}(X \le x)$ can be different than $\mathrm{P}(X < x)$.
- Even though any specific value of a continuous random variable has probability 0, *intervals* still can have positive probability.
- For continuous random variables, it doesn't really make sense to talk about the probability that the random value is *equal to* a particular value. In practical applications involving continuous random variables, "equal to" really means "close to", and "close to" probabilities correspond to intervals which can have positive probability.

- The density $f_X(x)$ at value x is *not* a probability.
- Rather, the density $f_X(x)$ at value x is related to the probability that the RV X takes a value "close to x" in the following sense

$$\mathrm{P}\left(x - \frac{\epsilon}{2} \le X \le x + \frac{\epsilon}{2}\right) \approx f_X(x)\epsilon, \qquad \text{for small } \epsilon$$

- The quantity ϵ is a small number that represents the desired degree of precision. For example, rounding to two decimal places corresponds to $\epsilon = 0.01$.
- Roughly,

$$\mathrm{P}(X \text{ is close to } x) \approx f_X(x) \times (\text{what counts as close to})$$

- What's important about a pdf is *relative* heights. For example, if $f_X(x_2) = 2f_X(x_1)$ then X is roughly "twice as likely to be close to x_2 than to be close to x_1" in the above sense.

$$\frac{f_X(x_2)}{f_X(x_1)} = \frac{f_X(x_2)\epsilon}{f_X(x_1)\epsilon} \approx \frac{\mathrm{P}\left(x_2 - \frac{\epsilon}{2} \le X \le x_2 + \frac{\epsilon}{2}\right)}{\mathrm{P}\left(x_1 - \frac{\epsilon}{2} \le X \le x_1 + \frac{\epsilon}{2}\right)} \approx \frac{\mathrm{P}(X \text{ is close to } x_2)}{\mathrm{P}(X \text{ is close to } x_1)}$$

Example 12.3. Suppose that X is a claim amount in excess of the decuctible for a car insurance policy that incurs a claim, measured in thousands of dollars. Assume that X has pdf

$$f_X(x) = (1/4.3)e^{-x/4.3}, \qquad x > 0$$

1. Sketch the pdf of X and interpret, roughly, its shape in context.

2. Without integrating, approximate the probability that X rounded to the nearest hundred dollars is \$500.

3. Compute and interpret the ratio $f(0.5)/f(5)$.

4. Compute and interpret $\mathrm{P}(X < 5)$.

5. Compute and interpret $\mathrm{P}(X > 10)$.

12.1 Exercises

Exercise 12.1. In a certain population, household income X (\$ thousands) follows the pdf

$$f_X(x) = cx^{-2.5}, \quad x \geq 30$$

for an appropriate constant c. Note that the measurement units are \$ thousands, so 1 represents 1 thousand dollars. (This is not a super realistic example, but just go with it.)

1. What are the possible values of X? Sketch the pdf of X.

2. Without integrating, find the probability that a household has an income, rounded to the nearest thousand dollars, of \$200K.

3. Without integrating, find the probability that a household has an income, rounded to the nearest *hundred* dollars, of \$200K.

4. Without integrating, determine how many times more likely it is for a household to have an income, rounded to the nearest *hundred* dollars, of \$100K than \$200K.

5. Compute the value of c. (For this and the following parts, set up any calculations you would need to perform, but you can use a calculator or software to do the computations.)

6. Compute the probability that a household has an income under \$100K.

7. Compute the probability that a household has an income over \$200K.

8. Find the probability that a household has an income exactly equal to \$200K.

Exercise 12.2. In Example 12.1, assume instead that employees are "equally likely" to arrive at any time between 8am and 9am. Let X be the arrival time of a randomly selected employee, measured in minutes after 8am. We'll assume that X has a "Uniform distribution" on [0, 60].

1. What shape would you want the pdf to have? Sketch that shape and then determine the pdf.

2. Compute and interpret $\mathrm{P}(X < 15)$.

3. Compute and interpret $\mathrm{P}(X > 45)$.

4. How do the two previous parts compare to each other? Why? How do they compare to the corresponding parts in Example 12.1? Why?

5. Compute $\mathrm{P}(X = 15)$.

6. Replace the previous part with a more sensible question, then solve it.

7. Make an educated guess for $\mathrm{E}(X)$.

8. Make an educated guess for $\mathrm{SD}(X)$. (We'll see how to compute expected value and variance for continuous random variables later, but for this example you should be able to make reasonable guesses without doing much computation.)

- A continuous random variable X has a **Uniform distribution** with parameters a and b, with $a < b$, if its probability density function f_X satisfies

$$
\begin{aligned}
f_X(x) &\propto \text{constant}, && a < x < b && (12.4)\\
&= \frac{1}{b-a}, && a < x < b. && (12.5)
\end{aligned}
$$

- If X has a Uniform(a, b) distribution then

$$
\begin{aligned}
\mathrm{E}(X) &= \frac{a+b}{2} && (12.6)\\
\mathrm{Var}(X) &= \frac{|b-a|^2}{12} && (12.7)\\
\mathrm{SD}(X) &= \frac{|b-a|}{\sqrt{12}} \approx 0.289|b-a| && (12.8)
\end{aligned}
$$